Area 1: Basic Engineering Practice

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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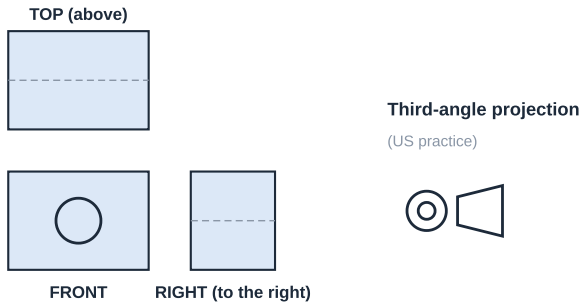
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1.A Engineering Terms, Symbols, and Drawings

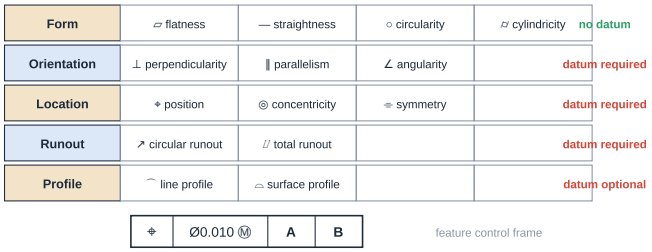
1. Key concepts

- Orthographic views: US practice is third-angle projection. The top view sits above the front view and the right-side view sits to its right. First-angle (ISO-E) mirrors that layout.
- Section views show internal features. Full section: the cutting plane goes all the way through and the cut surface is hatched. Half section: one quarter is removed. Broken-out, revolved and removed sections are local variants.
- Dimension notation: a boxed dimension is BASIC (theoretically exact; the tolerance comes from a feature control frame). A dimension in parentheses is REFERENCE (information only, not inspected).
- GD&T (ASME Y14.5) has five families. Form (flatness, straightness, circularity, cylindricity) never takes a datum. Orientation (perpendicularity, parallelism, angularity), location (position, concentricity, symmetry) and runout (circular, total) need datums. Profile can go either way.
- Material conditions: MMC is the condition with the most material (smallest hole, largest shaft). LMC is the opposite. RFS is the default. A bonus tolerance applies when a feature departs from MMC and the MMC modifier is used.
- Virtual condition (VC) is the worst-case boundary. Shaft: VC = MMC + geometric tolerance. Hole: VC = MMC − geometric tolerance. VC sets the size of functional gauges and the mating clearance.

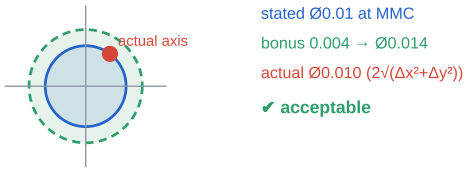
Diagrams



Third-angle projection: top view above the front view, right view to its right.



The five GD&T families and whether each needs a datum.



Position tolerance zone with a bonus at LMC.

2. Essential formulas

Name	Formula	Variables / units
Bonus tolerance	$\text{Total position tol} = \text{stated tol} + \lvert\text{actual size} - \text{MMC size}\rvert$	valid only with the Ⓜ modifier; sizes in in or mm
Virtual condition (external feature)	$\text{VC} = \text{MMC} + \text{geometric tol}$	shaft/pin
Virtual condition (internal feature)	$\text{VC} = \text{MMC} - \text{geometric tol}$	hole

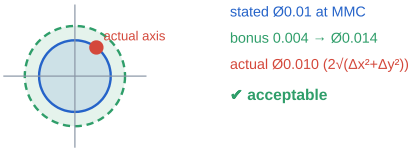
Position from coordinates	Diametral position error = $2\sqrt{(\Delta x^2 + \Delta y^2)}$	Δx, Δy = measured deviation from the true position

3. Handbook search keywords

third angle projection · section view · GD&T · position tolerance · maximum material condition · virtual condition · datum

4. Worked examples

Example 1: Bonus tolerance and position check
A Ø0.500 +0.006/−0.000 in hole has position tolerance Ø0.010 at MMC. A hole measures Ø0.504, and its axis is 0.003 in off in x and 0.004 in off in y. Is it acceptable?
Handbook search: **position tolerance**

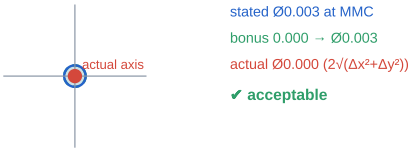


1. MMC of the hole = smallest size = 0.500 in.
2. Bonus = 0.504 − 0.500 = 0.004 in, so the allowed position = 0.010 + 0.004 = 0.014 in (diameter zone).
3. Actual diametral error = 2√(0.003² + 0.004²) = 0.010 in.
4. 0.010 ≤ 0.014, so the hole is acceptable.

Answer: Acceptable (0.010 in used of 0.014 in allowed)

Check your answer: The diametral error (0.010) must be no larger than the total allowed zone (0.014). Both are diameters.

Example 2: Virtual condition of a pin
A pin is Ø0.750 ±0.002 in with perpendicularity Ø0.003 at MMC. Find its virtual condition.
Handbook search: **virtual condition**

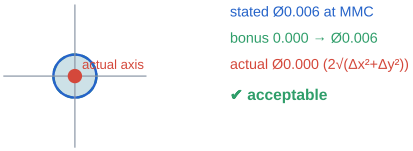


1. MMC of the pin = largest size = 0.752 in.
2. VC = MMC + tol = 0.752 + 0.003 = 0.755 in.

Answer: 0.755 in

Check your answer: A pin's VC is always bigger than its largest size; a hole's VC is smaller than its smallest size.

Try it yourself: Virtual condition of a hole
A hole is Ø1.000 +0.004/−0.000 in with position Ø0.006 at MMC. Find the virtual condition and the largest pin that always fits.
Handbook search: **virtual condition**



1. MMC of the hole = smallest size = 1.000 in.
2. VC = MMC − tol = 1.000 − 0.006 = 0.994 in.
3. A pin no larger than Ø0.994 in always assembles.

Answer: VC = 0.994 in

Check your answer: Hole VC must come out SMALLER than the hole's MMC. If you got 1.006, you used the shaft rule.

5. Common traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation √(Δx²+Δy²) instead of the diametral 2√(Δx²+Δy²).
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).
- Mixing up first-angle and third-angle view placement.

1.A Engineering Terms, Symbols, and Drawings: one-page cheat summary

- ▶ 3rd angle: top view above, right view to the right
- ▶ Boxed = basic; (x) = reference
- ▶ Form tolerances take no datum
- ▶ Hole MMC = smallest; shaft MMC = largest
- ▶ Bonus = |actual − MMC| (Ⓜ only)
- ▶ Position error = $2\sqrt{(\Delta x^2+\Delta y^2)}$
- ▶ VC shaft = MMC + tol; VC hole = MMC − tol

Formulas

Bonus tolerance	Total position tol = stated tol + |actual size − MMC size|
Virtual condition (external feature)	VC = MMC + geometric tol
Virtual condition (internal feature)	VC = MMC − geometric tol
Position from coordinates	Diametral position error = $2\sqrt{(\Delta x^2 + \Delta y^2)}$

Search keywords

third angle projection, section view, GD&T, position tolerance, maximum material condition, virtual condition, datum

Top traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation $\sqrt{(\Delta x^2+\Delta y^2)}$ instead of the diametral $2\sqrt{(\Delta x^2+\Delta y^2)}$.
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).

1.B Project Management and Economic Analysis

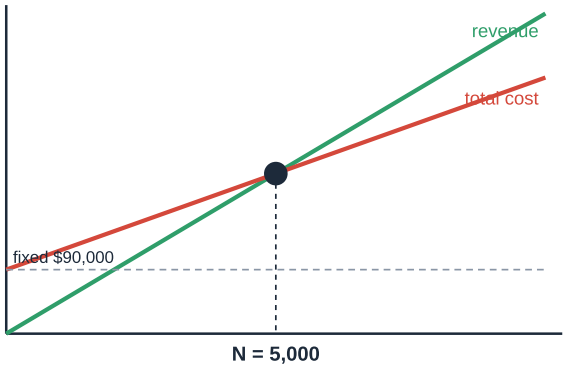
1. Key concepts

- Time value of money: move every cash flow to the same point in time (P, F or A) at the interest rate i before you compare.
- Mutually exclusive alternatives: pick the best PW, AW or EUAC. With unequal lives, use annual worth, or PW over the least common multiple of the lives.
- Sunk costs are irrelevant. Only future cash flows that differ between alternatives count.
- Break-even: the volume at which revenue equals total cost. Payback ignores the time value of money and anything after payback.
- Depreciation: straight line D = (P − S)/n. MACRS uses table percentages. Book value = cost − accumulated depreciation.
- Project scheduling (CPM): the critical path is the LONGEST path through the network, and its activities have zero total float. Earned value: CPI = EV/AC, SPI = EV/PV (above 1 is good).

Diagrams



Cash-flow diagram: cost at time 0, uniform savings, salvage at the end.



Break-even: revenue line crosses total cost.

2. Essential formulas

Name	Formula	Variables / units
Single payment	$F = P(1+i)^{n}$; $P = F(1+i)^{-n}$	i per period, n periods
Uniform series PW	$P = A[(1+i)^{n}-1]/[i(1+i)^{n}]$	A at the end of each period
Capital recovery	$A = P{\cdot}i(1+i)^{n}/[(1+i)^{n}-1]$	(A/P,i,n)
Sinking fund	$A = F{\cdot}i/[(1+i)^{n}-1]$	(A/F,i,n)
EUAC	$EUAC = P(A/P) - S(A/F) + \text{annual O\&M}$	S = salvage
Break-even	$N = \text{Fixed cost}/(\text{price} - \text{variable cost per unit})$	
Effective rate	$i_eff = (1 + r/m)\text{^}m - 1$	r nominal, m compoundings/yr
Earned value	$CPI = EV/AC$; $SPI = EV/PV$	

3. Handbook search keywords

interest factor tables · capital recovery · sinking fund · equivalent uniform annual cost · break-even · depreciation · critical path

4. Worked examples

Example 1: EUAC of a machine
A machine costs $50,000, lasts 10 years, has a $8,000 salvage value and $3,000/yr O&M. At i = 8%, find its EUAC.
Handbook search: **capital recovery**

1. (A/P,8%,10) = 0.14903; (A/F,8%,10) = 0.06903.
2. EUAC = 50,000(0.14903) − 8,000(0.06903) + 3,000 = $9,899/yr.



Answer: ≈ $9,899/yr

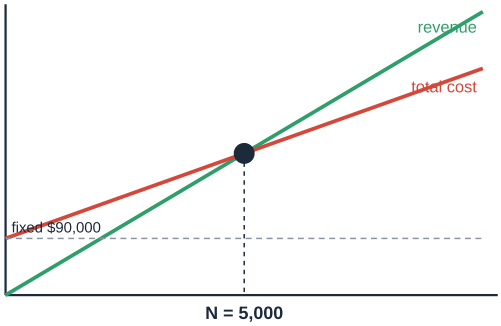
Check your answer: EUAC should be a bit more than 50,000/10 + 3,000 = 8,000 because interest adds cost, and salvage lowers it slightly.

Example 2: Break-even volume
Fixed cost $90,000/yr, price $45/unit, variable cost $27/unit. Find the break-even volume.
Handbook search: **break-even**

1. N = 90,000/(45 − 27) = 5,000 units/yr.

Answer: 5,000 units/yr

Check your answer: At 5,000 units revenue = 225,000 and cost = 90,000 + 135,000 = 225,000. They match.



Example 3: Present worth of savings
An upgrade saves $11,000/yr for 20 years. At 8%, what is the present worth?
Handbook search: **uniform series present worth**

1. (P/A,8%,20) = $[(1.08)^{20} − 1]/[0.08(1.08)^{20}]$ = 9.8181.
2. P = 11,000 × 9.8181 = $108,000.

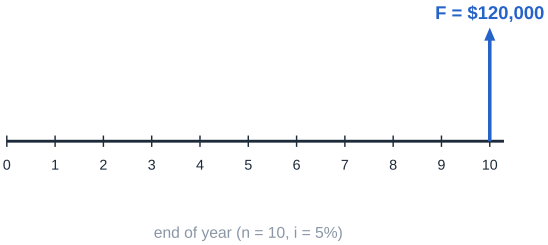


Answer: ≈ $108,000

Check your answer: PW must be less than the undiscounted total of 11,000 × 20 = 220,000.

Try it yourself: Sinking fund deposit
You need $120,000 in 10 years. At 5%, what end-of-year deposit is required?
Handbook search: **sinking fund**

1. (A/F,5%,10) = $0.05/[(1.05)^{10} − 1]$ = 0.07950.
2. A = 120,000 × 0.07950 = $9,541/yr.

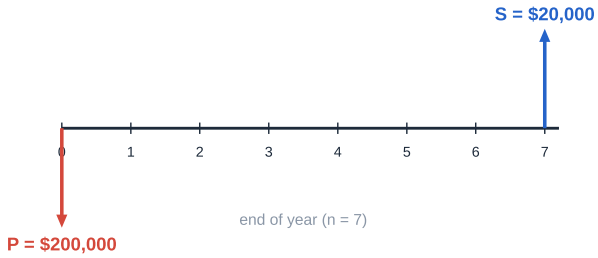


Answer: ≈ $9,541/yr

Check your answer: The deposit must be less than 120,000/10 = 12,000 because interest does part of the work.

Example 5: Straight-line book value
Equipment costs $200,000, life 7 years, salvage $20,000.
Find the book value after year 3.
Handbook search: **depreciation straight line**

1. D = (200,000 − 20,000)/7 = $25,714/yr.
2. BV_3 = 200,000 − 3 × 25,714 = $122,857.



Answer: $122,857

Check your answer: Book value must sit between cost (200,000) and salvage (20,000), and fall in a straight line.

5. Common traps

- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.
- Including sunk costs in a replacement study.
- Calling the path with the most activities the critical path (it is the longest in time).

1.B Project Management and Economic Analysis: one-page cheat summary

- ► $(A/P) = i(1+i)^n/[(1+i)^n-1]$
- ► $(A/F) = i/[(1+i)^n-1] = (A/P) - i$
- ► $(P/A) = 1/(A/P)$
- ► $EUAC = P(A/P) - S(A/F) + O\&M$
- ► Break-even $N = FC/(p - v)$
- ► SL depreciation $(P-S)/n$
- ► Critical path = longest, zero float

Formulas

Single payment	$F = P(1+i)^n$; $P = F(1+i)^{-n}$
Uniform series PW	$P = A[(1+i)^n-1]/[i(1+i)^n]$
Capital recovery	$A = P{\cdot}i(1+i)^n/[(1+i)^n-1]$
Sinking fund	$A = F{\cdot}i/[(1+i)^n-1]$
EUAC	EUAC = P(A/P) − S(A/F) + annual O&M
Break-even	N = Fixed cost/(price − variable cost per unit)
Effective rate	i_eff = (1 + r/m)^m − 1
Earned value	CPI = EV/AC ; SPI = EV/PV

Search keywords

interest factor tables, capital recovery, sinking fund, equivalent uniform annual cost, break-even, depreciation, critical path

Top traps

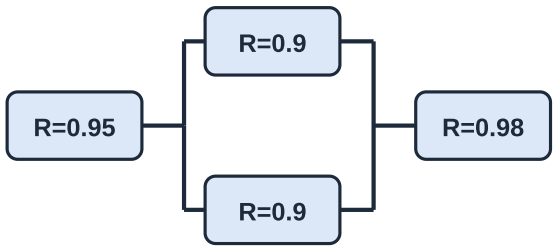
- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.

1.C Design Methodology

1. Key concepts

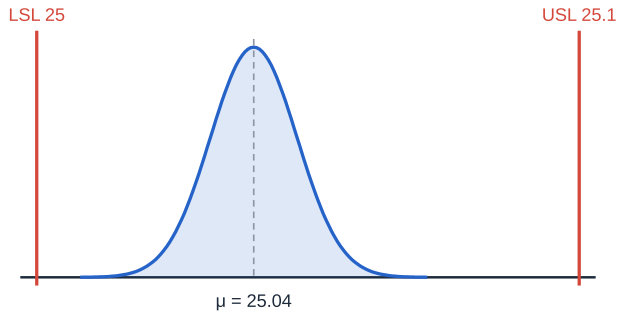
- The design process runs: define requirements → generate concepts → analyze and select (decision matrix) → detail design → prototype and test → release. It iterates.
- Verification asks 'did we build it right?' (does it meet the spec). Validation asks 'did we build the right thing?' (does it meet the user's need).
- FMEA rates each failure mode for Severity, Occurrence and Detection (1–10). RPN = S×O×D ranks the actions. A high severity alone also gets attention.
- Factor of safety n = strength/stress (or capacity/load). Reliability-based design uses distributions instead of a single n.
- Reliability: series $R = \Pi R_i$; parallel (redundant) $R = 1 - \Pi(1 - R_i)$. Constant failure rate: $R(t) = e^{(-\lambda t)}$, MTBF = $1/\lambda$.
- Root-cause failure analysis: preserve the evidence → fractography (beach marks show fatigue, a cup-cone shows ductile overload) → root cause → corrective action → verify.

Diagrams



R_sys = 0.9217

Series-parallel reliability block diagram.



σ = 0.008; Cpk = 1.67

Process spread versus specification limits.

2. Essential formulas

Name	Formula	Variables / units
Factor of safety	$n = S/\sigma$ (or allowable load/actual load)	
Series reliability	$R_s = R_1 R_2 \ldots R_n$	
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$	
Exponential reliability	$R(t) = e^{(-\lambda t)}$; MTBF = $1/\lambda$	λ in failures/h
FMEA	$RPN = S \times O \times D$	each 1–10

3. Handbook search keywords

factor of safety · reliability series parallel · FMEA · failure rate · design process

4. Worked examples

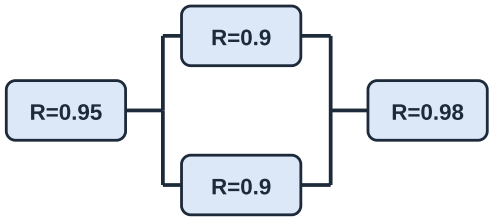
Example 1: System reliability
Pump A (R = 0.95) feeds two redundant valves in parallel (each R = 0.90), then a controller (R = 0.98). Find the system reliability.

Handbook search: **reliability series parallel**

1. Valve pair: $1 - (0.10)^2 = 0.99$.
2. System: $0.95 \times 0.99 \times 0.98 = 0.9217$.

Answer: 0.922

Check your answer: System reliability can never exceed the weakest series block (0.95).



R_sys = 0.9217

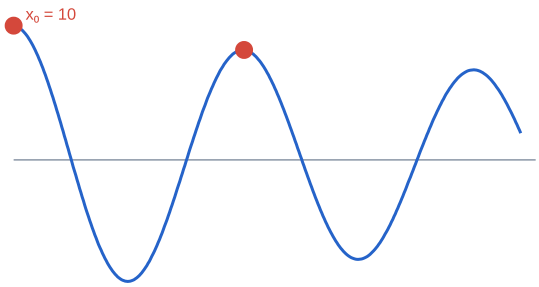
Example 2: Exponential life
A component has $\lambda = 2 \times 10^{-5}$ failures/h. Find R at 10,000 h and the MTBF.

Handbook search: **failure rate**

1. R = e^(−0.2) = 0.8187.
2. MTBF = 1/λ = 50,000 h.

Answer: R ≈ 0.819; MTBF = 50,000 h

Check your answer: R(t) at t = MTBF/5 should be e^(−0.2) ≈ 0.82.



$\delta = (1/n)\ln(x_0/x_n)$, n = 1 cycles

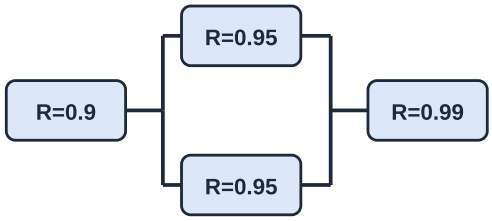
Try it yourself: Redundant pair in series
A (R = 0.9) is in series with two parallel B units (each 0.95) and then C (R = 0.99). Find R_sys.

Handbook search: **reliability series parallel**

1. Parallel B: $1 - 0.05^2 = 0.9975$.
2. R_sys = $0.9 \times 0.9975 \times 0.99 = 0.8888$.

Answer: 0.889

Check your answer: Below 0.9 (block A), as it must be for a series chain.



R_sys = 0.8888

5. Common traps

- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.C Design Methodology: one-page cheat summary

- ► Series: multiply R
- ► Parallel: $1 - \Pi(1-R)$
- ► $R = e^{(-\lambda t)}$, $MTBF = 1/\lambda$
- ► RPN = S·O·D
- ► n = strength/stress
- ► Verify = spec; validate = user need

Formulas

Factor of safety	$n = S/\sigma$ (or allowable load/actual load)
Series reliability	$R_s = R_1 R_2 \ldots R_n$
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$
Exponential reliability	$R(t) = e^{(-\lambda t)}$; $MTBF = 1/\lambda$
FMEA	$RPN = S \times O \times D$

Search keywords

factor of safety, reliability series parallel, FMEA, failure rate, design process

Top traps

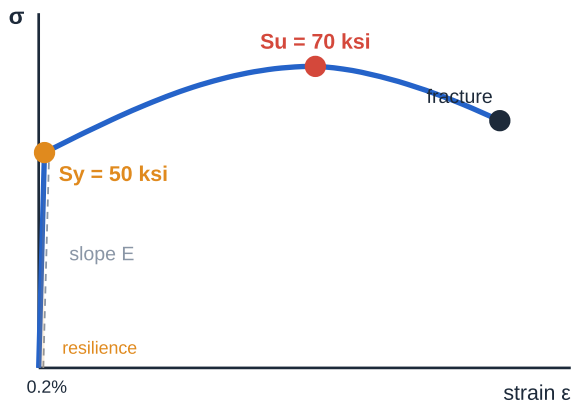
- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.D Physical Properties of Materials

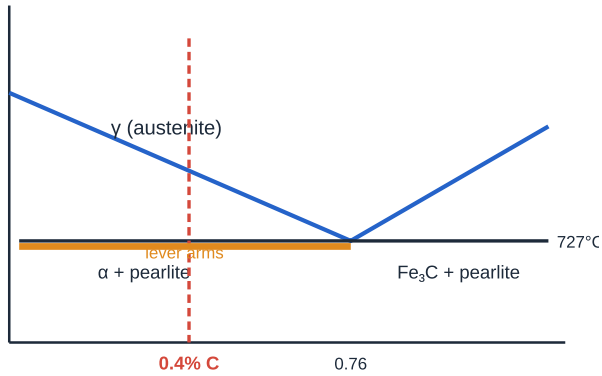
1. Key concepts

- Tension test: the elastic slope is E, then the 0.2% offset gives yield Sy, the peak gives the ultimate Su, then necking and fracture. Ductility is measured by %elongation or %reduction of area.
- True stress $\sigma_T = \sigma(1+\varepsilon)$ and true strain $\varepsilon_T = \ln(1+\varepsilon)$ hold up to necking (constant volume). Strain hardening: $\sigma_T = K\varepsilon_T^n$, and necking starts at $\varepsilon_T = n$.
- Resilience (elastic energy per volume) $u_r = Sy^2/2E$. Toughness is the area under the whole curve, approximately $(Sy+Su)/2 \cdot \varepsilon_f$.
- Hardness tracks strength for steels: $Su \approx 0.5 \cdot HB$ ksi ($3.45 \cdot HB$ MPa).
- Fe–C diagram: eutectoid 0.76% C at 727 °C (pearlite). Hypoeutectoid steels hold proeutectoid ferrite plus pearlite; use the lever rule. Quenching gives martensite and tempering restores toughness.
- BCC metals (ferritic steels) have a ductile-to-brittle transition. FCC metals (Al, Cu, austenitic stainless) do not. Creep matters above about 0.4 T_m (absolute); the Larson–Miller parameter is $LMP = T(C + \log t)$.
- Thermal expansion: $\delta = \alpha L \Delta T$. Galvanic corrosion attacks the anode (e.g. steel against copper); avoid a small anode with a large cathode.
- Composites: the rule of mixtures $E_L = E_f V_f + E_m(1-V_f)$ applies along the fibers (iso-strain). Across the fibers (iso-stress): $1/E_T = V_f/E_f + V_m/E_m$.

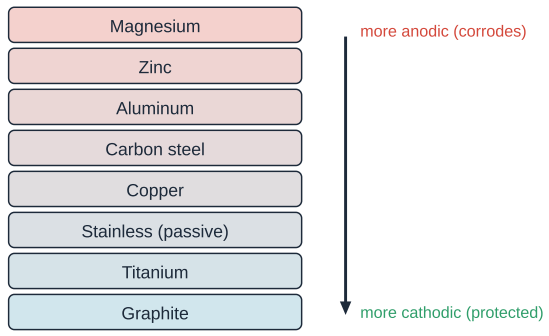
Diagrams



Engineering stress-strain curve: yield (0.2% offset), ultimate, fracture.



Simplified Fe–Fe_3C diagram with the lever rule at 0.4% C.



Galvanic series in seawater.

2. Essential formulas

Name	Formula	Variables / units
Hooke	$\sigma = E\varepsilon$; $\varepsilon = \delta/L$	E: psi or GPa

Poisson	$\varepsilon_{lat} = -\nu\varepsilon_{axial}$	$\nu \approx 0.3$ for steel
True stress/strain	$\sigma_T = \sigma(1+\varepsilon);\ \varepsilon_T = \ln(1+\varepsilon)$	before necking
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$	
Resilience	$u_r = Sy^2/(2E)$	in-lb/in³
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$	wt% C
Larson–Miller	$LMP = T(C + \log t)$	T absolute (°R or K), t in h, C ≈ 20
Thermal strain	$\delta = \alpha L \Delta T$	α per °F or °C
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$	

3. Handbook search keywords

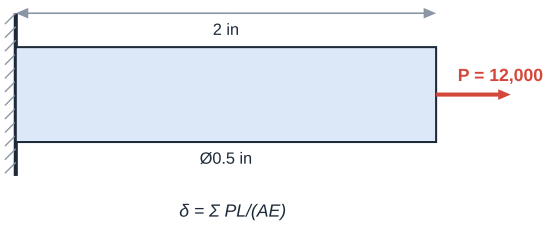
stress-strain diagram · true stress · modulus of resilience · hardness conversion · iron-carbon phase diagram · lever rule · Larson-Miller · coefficient of thermal expansion · galvanic series

4. Worked examples

Example 1: Tension test reduction

A 0.500 in diameter steel bar ($E = 30\times10^6$ psi, $\nu = 0.29$) carries 12,000 lb. Find the stress, the axial strain and the change in diameter.

Handbook search: **Poisson's ratio**



1. $A = \pi(0.5)^2/4 = 0.1963$ in²; $\sigma = 12{,}000/0.1963 = 61{,}115$ psi.
2. $\varepsilon = \sigma/E = 2.037\times10^{-3}$.
3. $\Delta d = -\nu\cdot\varepsilon\cdot d = -0.29 \times 2.037\times10^{-3} \times 0.5 = -2.95\times10^{-4}$ in (it contracts).

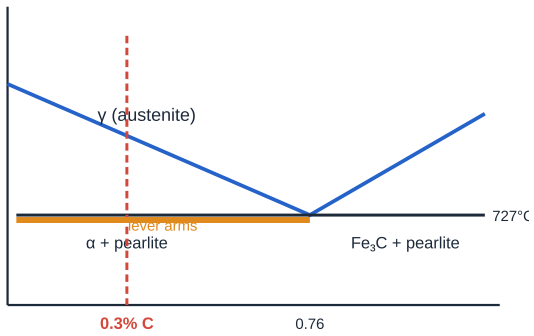
Answer: $\sigma \approx 61{,}115$ psi; $\Delta d \approx 295$ µin decrease

Check your answer: Lateral strain is negative and about 0.3 × the axial strain.

Example 2: Lever rule

A 0.30% C steel is slowly cooled to just below 727 °C (α at 0.022% C, eutectoid at 0.76% C). Find the fraction of proeutectoid ferrite.

Handbook search: **lever rule**



1. $W_{\alpha'} = (0.76 - 0.30)/(0.76 - 0.022) = 0.623$.
2. Pearlite = 0.377.

Answer: 0.623

Check your answer: Less carbon means more proeutectoid ferrite. 0.30% C should give well over half ferrite.

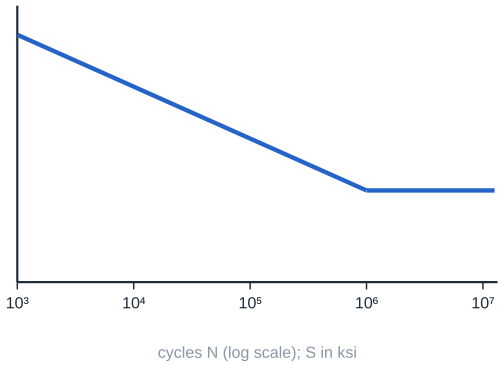
Example 3: Larson–Miller
LMP = 38,000 (°R, h, C = 20). Find the rupture life at 1,200 °F.

Handbook search: **Larson-Miller**

1. T = 1,200 + 460 = 1,660 °R.
2. log t = 38,000/1,660 − 20 = 2.892 → t = 779 h.

Answer: ≈ 779 h

Check your answer: A higher temperature always gives a shorter life for the same LMP.



cycles N (log scale); S in ksi

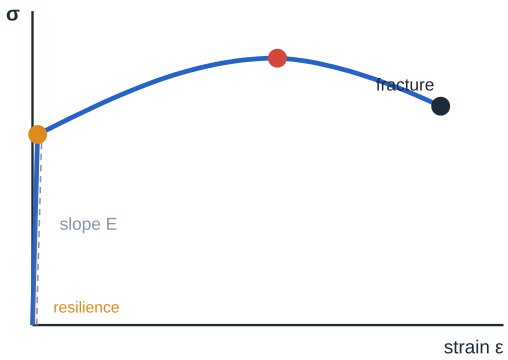
Try it yourself: Modulus from a tension test
A Ø0.6 in specimen with 2.000 in gage length stretches 0.008 in under 26,000 lb (elastic). Find E.

Handbook search: **modulus of elasticity**

1. σ = P/A = 26,000/(π·0.6²/4) = 91,956 psi.
2. ε = 0.008/2.000 = 0.004.
3. E = σ/ε = 22,989,047 psi ≈ 23.0×10⁶ psi.

Answer: ≈ 23.0×10⁶ psi

Check your answer: Steel is about 29–30×10⁶ psi. A result far from that points to a units or diameter-vs-radius slip.



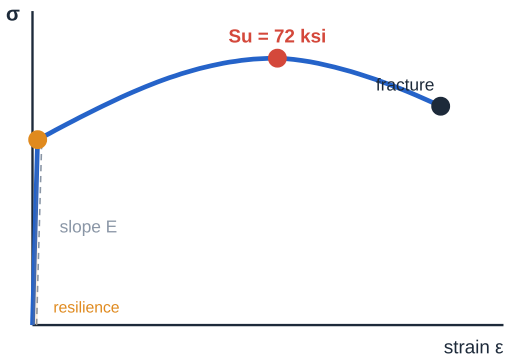
Example 5: True stress
Engineering stress 72,000 psi at engineering strain 0.15 (before necking). Find the true stress and true strain.

Handbook search: **true stress**

1. σ_T = σ(1 + ε) = 72,000 × 1.15 = 82,800 psi.
2. ε_T = ln(1 + ε) = 0.1398.

Answer: σ_T ≈ 82,800 psi; ε_T ≈ 0.140

Check your answer: True stress is always higher than engineering stress in tension, and true strain a little lower.



5. Common traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with σ(1+ε) (the formula is invalid there).
- Putting the lever-rule numerator on the wrong side (use the OPPOSITE arm).
- Expecting FCC metals to show a DBTT.

1.D Physical Properties of Materials: one-page cheat summary

- ► $\sigma = P/A, \varepsilon = \delta/L, E = \sigma/\varepsilon$
- ► $\sigma_T = \sigma(1+\varepsilon), \varepsilon_T = \ln(1+\varepsilon)$
- ► $\%RA = 1 - (d_f/d_0)^2$
- ► $u_r = Sy^2/2E$
- ► $Su \approx 0.5\ HB\ ksi$
- ► Eutectoid 0.76% C, 727 °C
- ► $LMP = T_{abs}(20 + \log t)$
- ► $\delta = \alpha L \Delta T$
- ► $E_L = E_f V_f + E_m V_m$

Formulas

Hooke	$\sigma = E\varepsilon\ ;\ \varepsilon = \delta/L$
Poisson	$\varepsilon_{lat} = -\nu\varepsilon_{axial}$
True stress/strain	$\sigma_T = \sigma(1+\varepsilon);\ \varepsilon_T = \ln(1+\varepsilon)$
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$
Resilience	$u_r = Sy^2/(2E)$
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$
Larson–Miller	$LMP = T(C + \log t)$
Thermal strain	$\delta = \alpha L \Delta T$
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$

Search keywords

stress-strain diagram, true stress, modulus of resilience, hardness conversion, iron-carbon phase diagram, lever rule, Larson-Miller, coefficient of thermal expansion, galvanic series

Top traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with σ(1+ε) (the formula is invalid there).